

Computer Graphics - Question 3

Answer Any Eight | 2 Marks Each | Theory + Clear Mathematics

3 (a) What do you mean by persistence?

Persistence is the time for which the phosphor of a CRT screen continues to emit light after the electron beam is removed. It is generally measured as the time required for the emitted light intensity to decrease to one-tenth of its original value.

3 (b) Define resolution and aspect ratio.

Resolution is the number of pixels that can be displayed on a screen, for example 1920 x 1080. **Aspect ratio** is the ratio of the width of a display or image to its height, for example 16:9.

3 (c) Write the basic procedure of DDA line drawing algorithm.

DDA (Digital Differential Analyzer) is a line drawing algorithm that calculates intermediate points between two endpoints. For endpoints (x1, y1) and (x2, y2), first calculate dx and dy, then find the number of steps and coordinate increments.

$$dx = x2 - x1 \quad dy = y2 - y1$$

$$\text{Steps} = \max(|dx|, |dy|)$$

$$Xinc = dx / \text{Steps} \quad Yinc = dy / \text{Steps}$$

$$\text{Next point: } x = x + Xinc, \quad y = y + Yinc$$

3 (d) Write the formula for condition checking in ellipse.

An ellipse is represented using horizontal radius rx and vertical radius ry. In the midpoint ellipse algorithm, the following condition is used to determine the change from Region 1 to Region 2.

Ellipse equation:

$$x^2 \quad y^2$$

$$----- + ----- = 1$$

$$rx^2 \quad ry^2$$

$$\text{Region condition: } 2(ry^2)x \geq 2(rx^2)y$$

3 (e) Write the formula for midpoint circle algorithm.

The midpoint circle algorithm draws a circle by choosing the nearest pixel using a decision parameter. For a circle of radius r, the initial and updating formulas are:

$$P_0 = 1 - r$$

If $P_k < 0$:

$$P_{k+1} = P_k + 2x_k + 3$$

Otherwise:

$$P_{k+1} = P_k + 2(x_k - y_k) + 5$$

3 (f) Explain translation with matrix representation.

Translation is a 2-D transformation used to move an object from one position to another without changing its size, shape or orientation. If Tx and Ty are the translation distances along the X and Y axes, then:

$$x' = x + Tx \quad y' = y + Ty$$

$$\begin{array}{rcccl} x' & & 1 & 0 & T_x & x \\ y' & = & 0 & 1 & T_y & y \\ 1 & & 0 & 0 & 1 & 1 \end{array}$$

3 (g) Explain rotation with matrix representation.

Rotation is a 2-D transformation that turns an object through an angle theta about a fixed point. For rotation about the origin in the anticlockwise direction, the equations are:

$$x' = x \cos(\theta) - y \sin(\theta)$$

$$y' = x \sin(\theta) + y \cos(\theta)$$

$$\begin{array}{rcccl} x' & & \cos(\theta) & -\sin(\theta) & 0 & x \\ y' & = & \sin(\theta) & \cos(\theta) & 0 & y \\ 1 & & 0 & 0 & 1 & 1 \end{array}$$

3 (h) What is point clipping?

Point clipping is the process of checking whether a point lies inside or outside a clipping window. A point P(x, y) is accepted only when its X and Y coordinates lie within the minimum and maximum boundaries of the window.

$$x_{min} \leq x \leq x_{max}$$

$$y_{min} \leq y \leq y_{max}$$

If both conditions are true -> point is accepted.

Otherwise -> point is rejected.